

SOVEREIGN INDUSTRIAL PLANT INTEGRITY MANAGEMENT
EQUIPMENT INSPECTION REPORT - ROTATING MACHINERY
REPORT NO: IR-2026-P202-0822
DATE OF INSPECTION: 2026-08-22
INSPECTION TYPE: Routine In-Service Reliability Audit

1. EQUIPMENT IDENTIFICATION

Equipment Tag: P-202
Equipment Description: Boiler Feed Water Secondary Booster Pump
Service: Demineralized Boiler Feed Water (Density: 998 kg/m3)
Drive Unit: 3.3kV High-Voltage Induction Motor (110 kW, 2950 RPM)
Governing Specifications: ISO 10816-3 Class II / SOP-M-104

2. OPERATIONAL MEASUREMENTS & SENSOR TELEMETRY

- Vibration (Drive End Bearing RMS): 5.1 mm/s (WARNING: ISO 10816-3 Zone C band 4.5 - 7.1 mm/s)
- Vibration (Outboard Bearing RMS): 3.2 mm/s (NORMAL: Zone B band 1.8 - 4.5 mm/s)
- Bearing Temperature: 71.0 deg C (NORMAL: SOP-M-104 threshold <= 75.0 deg C)
- Mechanical Seal Dynamic Leakage: 2 drops/minute (NORMAL: SOP-M-104 threshold <= 5 drops/min)
- Suction Pressure: 0.85 MPa (Design normal)
- Discharge Pressure: 4.20 MPa (Design normal)
- Flow Rate: 82.0 m3/h (Design normal 80 - 85 m3/h)
- Differential Head: 115.0 meters

3. PHYSICAL & NON-DESTRUCTIVE EXAMINATION FINDINGS

- Plan 53A dual mechanical seal barrier fluid reservoir is pressurized and stable. Seepage rate is minor (2 drops/min).
- Drive-end bearing temperature is stable at 71.0 deg C. Vibration spectrum exhibits elevated 1X rotational unbalance peak at 5.1 mm/s.
- Pump volute and suction pipe show uniform external surface oxidation with zero localized pitting.
- Ultrasonic thickness survey confirms casing wall thickness of 26.4 mm (well above 19.8 mm required minimum).

4. RISK SEVERITY & RISK CLASSIFICATION

Overall Risk Severity: HIGH (PLANNED OVERHAUL REQUIRED)
Risk Justification: Vibration falls into Zone C (4.5 - 7.1 mm/s), indicating mechanical unbalance or coupling wear. Equipment does not require emergency trip but necessitates overhaul within 14 days per SOP-M-104.

5. INSPECTOR RECOMMENDATIONS & CORRECTIVE ACTIONS

- Schedule planned maintenance overhaul within 14 days per SOP-M-104 Section 3.
- Inspect motor-pump flexible coupling elastomeric insert for wear and backlash.
- Re-lubricate bearing cavities with ISO VG 46 synthetic grease.
- Monitor triaxial vibration daily to ensure levels do not escalate toward Zone D (> 7.1 mm/s).

Lead Inspector: M. S. Thorne, Plant Reliability Engineer
Signature: [Verified / Stamped: Plant Reliability Directorate]

PLANT RELIABILITY DIVISION
OVERHAUL IN 14 DAYS
AIR-GAP FIELD VERIFIED